SIMATS ENGINEERING
ASSESSMENT TOOL1

COURSE CODE: MLA0202 COURSE NAME: Fundamentals of Machine Learning

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COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***Kruthika Sre S / 192525360*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kruthika Sre S

Q1. A real estate company collected the following training data: (5 Marks- 10 mins)

House No	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a linear regression model of the form Price = $\beta_0 + \beta_1$(Area) + β_2(Bedrooms)

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Answer:

Using the least-squares normal equations on the 5 data points (n = 5, ΣArea = 6500, ΣBedrooms = 14, ΣPrice = 315, ΣArea² = 9,330,000, ΣBedrooms² = 42, ΣArea·Bedrooms = 19700, ΣArea·Price = 446500, ΣBedrooms·Price = 945), solving the three simultaneous equations gives:

$\beta_0 \approx 8.50, \beta_1 \approx 0.0425, \beta_2 \approx -0.280$

(i) Model: Price = 8.50 + 0.0425(Area) − 0.280(Bedrooms)

(ii) Regression coefficients: $\beta_0 = 8.50$, $\beta_1 = 0.0425$, $\beta_2 = -0.280$

(iii) Interpretation: β_0 (8.50) is the baseline price when Area and Bedrooms are both zero (a mathematical reference point, not practically meaningful). β_1 (0.0425) means price increases by about ₹0.0425 Lakh (≈ ₹4,250) for every additional sq.ft of area, holding bedrooms constant. β_2 (−0.280) means price decreases by about ₹0.280 Lakh for every extra bedroom, holding area constant — this counter-intuitive sign arises because Area and Bedrooms are highly correlated in this small sample (multicollinearity), so the model is splitting the same price effect between the two variables.

Q2. An email system uses the following feature values: (5 Marks- 10 mins)

Email	Word Freq. (Offer)	Word Freq. (Win)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Answer:

Inspecting the sum (Offer + Win) for each email: Spam cases give 5, 3, 6 (all ≥ 3); Non-spam cases give 1, 1 (both ≤ 1). A simple linear classifier that separates the two groups is:

$f(x) = 1 \cdot (\text{Offer}) + 1 \cdot (\text{Win}) - 2$

(i) Model: classify as Spam (1) if $f(x) = \text{Offer} + \text{Win} - 2 \geq 0$, else Not Spam (0).

(ii) Decision boundary: Offer + Win = 2 (the line separating the two classes).

(iii) New email: Offer = 2, Win = 1 $\rightarrow$ sum = 3, $f(x) = 3 - 2 = 1 \geq 0$.

Classification: Spam (1).

Q3. A medical dataset is given: (5 Marks -10 Mins)

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

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Answer:

(i) Prior probabilities: P(Disease = Yes) = 3/5 = 0.6, P(Disease = No) = 2/5 = 0.4

(ii) Conditional probabilities (from the 3 "Yes" and 2 "No" patients):

P(Fever=1 | Yes) = 3/3 = 1,

P(Headache=0 | Yes) = 1/3,

P(Fever=1 | No) = 0/2 = 0,

P(Headache=0 | No) = 1/2

(iii) Using Naïve Bayes: $P(\text{Yes} \mid F=1,H=0) \propto P(F=1|\text{Yes}) \cdot P(H=0|\text{Yes}) \cdot P(\text{Yes}) = 1 \times (1/3) \times 0.6 = 0.2$, and $P(\text{No} \mid F=1,H=0) \propto P(F=1|\text{No}) \cdot P(H=0|\text{No}) \cdot P(\text{No}) = 0 \times (1/2) \times 0.4 = 0$.

Normalising, P(Disease = Yes | Fever=1, Headache=0) = 0.2 / (0.2 + 0) = 1 (≈ 100%).

Final answer: the patient is almost certainly predicted to have the disease (probability ≈ 1), since no training patient without fever existed among the "No" class, driving its likelihood to zero (the classic zero-frequency case — in practice this is handled with Laplace/add-one smoothing).

Q4. Given the following dataset: (5 Marks -10 Mins)

Data Point	X1	X2	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct a discriminant function.

(ii) Determine the decision boundary separating the two classes.

(iii) Classify a new point (X1 = 3, X2 = 3).

Answer:

Class means: mean(C1) = (2.5, 3.5) [from (2,3),(3,4)]; mean(C2) = (5, 6) [from (5,6),(6,7),(4,5)].

(i) Minimum-distance discriminant function: $g(x) = (\text{mean}_{C1} - \text{mean}_{C2}) \cdot x + w_0$, giving $g(x) = -2.5 \cdot X1 - 2.5 \cdot X2 + 21.25$ (classify C1 if g(x) > 0, else C2).

(ii) Decision boundary: setting g(x) = 0 simplifies to X1 + X2 = 8.5 — the perpendicular bisector of the two class means. (Check: C1 points give sums 5 and 7, both < 8.5; C2 points give sums 11, 13, 9, all > 8.5 — consistent with the training data.)

(iii) New point (3, 3): X1 + X2 = 6, which is < 8.5.

Classification: C1.

Q5. A telecom company collected the following data: (5 Marks -10 Mins)

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

(i) Formulate the Bayesian logistic regression model.

(ii) Estimate model parameters using the dataset.

(iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Answer:

(i) Model formulation: $P(\text{Churn}=1 \mid x) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1(\text{Charges}) + \beta_2(\text{Tenure})$, with Gaussian priors $\beta_i \sim N(0, \sigma^2)$ placed on the coefficients (Bayesian/MAP formulation) and a Bernoulli likelihood for Churn given the data.

(ii) With only 5 data points, the MAP estimate is approximated here using the least-squares normal equations (equivalent to a linearised/first-order approximation of the logistic link). Solving gives:

$\beta_0 \approx 5.54, \beta_1 \approx -0.0033, \beta_2 \approx -0.287$

indicating Tenure is the dominant driver of churn in this sample (longer tenure → much lower churn), while Monthly Charges has only a weak, slightly negative independent effect once tenure is accounted for.

(iii) For Charges = 650, Tenure = 9: $z = 5.54 - 0.0033(650) - 0.287(9) = 5.54 - 2.17 - 2.59 \approx 0.78$. Passing this through the sigmoid gives an estimated churn probability ≈ 0.79 (79%), which is above the 0.5 threshold.

Prediction: the customer is likely to Churn (Churn = 1).

Q6. Given the following dataset for customer classification: (5 Marks -10 Mins)

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

For a new customer (Age = 32, Income = 50):

(i) Compute Euclidean distance from all points.

(ii) Classify using k = 3.

(iii) Determine the final class label.

Answer:

(i) Euclidean distances from (32, 50): to C1(25,40) = $\sqrt{(7^2 + 10^2)} = \sqrt{149} \approx 12.21$; to C2(30,60) = $\sqrt{(2^2 + 10^2)} = \sqrt{104} \approx 10.20$; to C3(35,80) = $\sqrt{(3^2 + 30^2)} = \sqrt{909} \approx 30.15$; to C4(40,20) = $\sqrt{(8^2 + 30^2)} = \sqrt{964} \approx 31.05$.

(ii) k = 3 nearest neighbours (smallest distances): C2 (10.20, Class A), C1 (12.21, Class A), C3 (30.15, Class B).

(iii) Majority vote among the 3 neighbours: Class A appears twice (C1, C2), Class B nce (C3).

Final class label: A.

Q7. Given the dataset: (5 Marks -10 Mins)

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) Compute Entropy of Play.

(ii) Calculate Information Gain for attribute Outlook.

(iii) Identify the root node of the decision tree.

Answer:

(i) Play has 2 "Yes" and 2 "No" out of 4: Entropy(Play) = $-(2/4)\log_2(2/4) - (2/4)\log_2(2/4) = 0.5 + 0.5 = 1$ bit.

(ii) Splitting by Outlook: Sunny → {No, No} (entropy = 0, pure); Overcast → {Yes} (entropy = 0, pure); Rain → {Yes} (entropy = 0, pure). Weighted entropy after split = $(2/4)(0) + (1/4)(0) + (1/4)(0) = 0$.

Information Gain(Outlook) = Entropy(Play) − Weighted Entropy = 1 − 0 = 1 bit.

(iii) Since Outlook gives the maximum possible information gain (1 bit) and perfectly separates the classes, Outlook is chosen as the root node of the decision tree.

Q8. Given data points: (5 Marks -10 Mins)

X1	X2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

(i) Plot the points.

(ii) Determine a linear separating hyperplane.

(iii) Find the equation of the decision boundary.

Answer:

(i) Plotting the 4 points shows (2,2) and (4,4) lying on the diagonal X1 = X2, while (2,4) and (4,2) lie on the opposite diagonal X1 + X2 = 6 — this is the classic XOR-type arrangement.

(ii) No linear separating hyperplane exists for this data. Proof: the centroid of class +1 is ((2+4)/2, (2+4)/2) = (3, 3), and the centroid of class −1 is ((2+4)/2, (4+2)/2) = (3, 3) — the two classes have the identical centroid, so no straight line can place all +1 points on one side and all −1 points on the other (this is the well-known XOR problem, not linearly separable in 2 dimensions).

(iii) A non-linear boundary is needed instead. Using the product feature $(X1 - 3)(X2 - 3)$: it equals +1 for (2,2) and (4,4), and −1 for (2,4) and (4,2). So the (non-linear) decision boundary is $(X1 - 3)(X2 - 3) = 0$, i.e. the pair of lines X1 = 3 and X2 = 3, with class +1 where $(X1-3)(X2-3) > 0$ and class −1 where $(X1-3)(X2-3) < 0$.

Q9. Given confusion matrices: (5 Marks -10 Mins)

Logistic Regression:

	Pred +	Pred -
Actual +	40	10
Actual -	5	45

Naïve Bayes:

	Pred +	Pred -
Actual +	35	15
Actual -	10	40

(i) Compute Accuracy, Precision, Recall for both models.

(ii) Compare results and identify the better model.

Answer:

Logistic Regression — TP=40, FN=10, FP=5, TN=45, Total=100:

Accuracy = (40+45)/100 = 0.85 (85%); Precision = 40/(40+5) = 0.889 (88.9%); Recall = 40/(40+10) = 0.80 (80%).

Naïve Bayes — TP=35, FN=15, FP=10, TN=40, Total=100:

Accuracy = (35+40)/100 = 0.75 (75%); Precision = 35/(35+10) = 0.778 (77.8%); Recall = 35/(35+15) = 0.70 (70%).

(ii) Comparison: Logistic Regression scores higher on all three metrics — Accuracy (85% vs 75%), Precision (88.9% vs 77.8%) and Recall (80% vs 70%).

Better model: Logistic Regression.

Q10. Given points: (5 Marks -10 Mins)

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids: C1 = (2,4), C2 = (8,9)

(i) Perform one iteration of K-means.

(ii) Assign clusters.

(iii) Compute new centroids.

Answer:

(i)–(ii) Distance of each point to the two initial centroids, and resulting assignment:

Point	Dist. to C1(2,4)	Dist. to C2(8,9)	Assigned Cluster
(2,4)	0	$\sqrt{61} \approx 7.81$	C1
(3,5)	$\sqrt{2} \approx 1.41$	$\sqrt{41} \approx 6.40$	C1
(8,9)	$\sqrt{61} \approx 7.81$	0	C2
(9,10)	$\sqrt{85} \approx 9.22$	$\sqrt{2} \approx 1.41$	C2

Clusters formed: Cluster 1 = {(2,4), (3,5)}; Cluster 2 = {(8,9), (9,10)}.

(iii) New centroids (mean of each cluster's points):

C1_new = ((2+3)/2, (4+5)/2) = (2.5, 4.5)

C2_new = ((8+9)/2, (9+10)/2) = (8.5, 9.5)

RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation